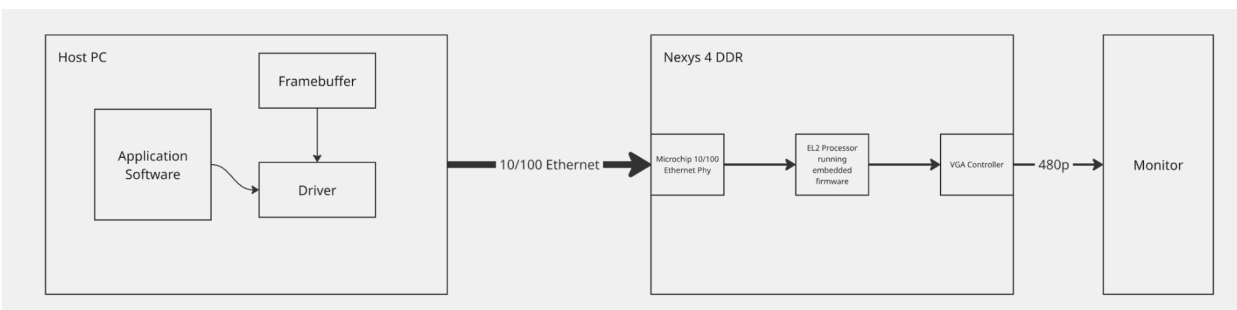
Final Project Proposal

Matthew Hardenburgh, Jose Ramirez, Paul Krueger

Introduction

The basic idea for this project is to stream the desktop of a host PC over ethernet to the Nexys board and then display it on a monitor in 480p. Essentially the Nexys board will act as a very rudimentary graphics card. The components of this project are, a desktop linux driver/userland application on the host PC that will connect to the board on the network and send it frames, firmware running on the Nexys board that will display those frames, and an ethernet controller peripheral designed in system Verilog. We will use the existing VGA controller from our previous project to display the frames. The challenge of this project will be designing the ethernet controller because it will operate at a faster clock than the rest of the board and we will be crossing clock domain. Using Ethernet is more difficult, but was chosen for this design because we wanted to use hardware that came with the Nexys board and not have to purchase new hardware.

Block Diagram



Stretch Goals

It would be a nice feature to play audio through the mono out port on the Nexys board. Given time, we may implement this feature